

Module 07 - Expenses & Splitting

Shared expense creation, split modes, balances

SOURCE OF TRUTH: This resource intentionally uses plain JavaScript. The locked stack is React + JavaScript + Firebase + Material UI (MUI). Do not introduce Next.js, TypeScript, shadcn/ui, Redux, or another backend framework unless the project owner explicitly changes this specification.

Scope

Implement expense CRUD and split calculation for equal, percentage, custom, and participant-based splits. Expenses belong to a group and can optionally reference a plan or booking.

Expense document

```
groups/{groupId}/expenses/{expenseId}
{
  planId,
  bookingId,
  description,
  amount,
  currency,
  paidBy,
  splitMode: "equal" | "percentage" | "custom" | "participants",
  createdBy,
  createdAt,
  updatedAt,
  status: "open" | "settled"
}
```

Split record

```
.../expenses/{expenseId}/splits/{uid}
{
  uid,
  owedAmount,
  paidAmountSnapshot,
  settledAmount,
  status
}
```

Split rules

Mode	Rule
Equal	amount / participantCount; handle currency rounding
Percentage	percentages must sum to 100
Custom	owed amounts must sum exactly to amount
Participants	Only selected participants share equally

Rounding

Money must never be represented as binary floating-point values in final persisted calculations. Use integer minor units (for INR, paise) for stored amounts and arithmetic where practical.

Flow

```
Create Expense
-> Validate amount + payer + participants
-> Calculate split
```

```
-> Persist expense + splits atomically where possible  
-> Create activity  
-> Update visible balance summaries
```

Acceptance criteria

- Split totals always reconcile to the original amount.
- Invalid percentage/custom splits are blocked.
- Expense can be edited only by permitted users.
- Settlement status is separate from expense deletion.
- Balance UI updates after expense creation without full page reload.